Acme Solar Technologies

Three statement forecasting and scenario analysis

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Case and modelling approach

Acme Solar Technologies (AST) is a fictional manufacturer of solar panels and energy-storage solutions serving residential, commercial and industrial customers. The case describes a vertically integrated business operating across North America and Europe, exposed to government incentives, material prices and customer demand.

The **Corporate Finance Institute (CFI)** supplied the fictional case, historical data and broad scenario framework. This project independently develops those materials into an integrated model, with project-specific forecasting assumptions, quantified scenario shocks, interpretation and recommendations.

The linked income statement, balance sheet and cash-flow model evaluates profitability, liquidity and financial flexibility over **2024-2026**. AST remains profitable and maintains positive quarter-end cash under the individual scenarios tested. However, adverse conditions materially reduce cash accumulation. Management should protect liquidity, maintain pricing discipline and evaluate investment commitments against updated cash requirements.

The model uses 12 historical quarters from 2021-2023. A driver-based forecast separates annual growth from quarterly seasonality and links working capital, capex, depreciation, debt and interest. Base revenue growth moderates to **10%, 9% and 8%**. Gross margin remains 40%, with operating expenses maintained at the 2023 revenue ratio.

Historical statements were reconciled before forecasting. Q1 2021 opening cash was inferred at $21.8m; an unexplained negative $0.2m opening equity adjustment remains separately disclosed. Neither is presented as a confirmed accounting correction.

Base forecast

USD million	2023 historical	2024	2025	2026
Revenue	950.0	1,045.0	1,139.1	1,230.2
Operating profit	180.0	198.0	215.8	233.1
Net income	120.0	134.8	150.2	165.3
Closing cash	210.8	226.8	265.8	323.2

Operating profit means operating income before interest and tax. The forecast is a planning exercise from the stated cutoff, rather than a description of subsequent realised events.

Scenario development

Three adverse cases isolate different sources of risk. Assumptions are **analyst judgements for this project**, rather than statistically estimated policy or market responses. Changes apply against the corresponding base quarter; they do not compound as repeated annual shocks. Slash-separated figures refer to **2024 / 2025 / 2026**.

Driver versus base	Policy shift	Cost volatility	Temporary demand
Trigger	Reduced support for solar customers	Higher material prices	Temporary subsidies favour competing energy
Sales volume	-10% throughout	Unchanged	-20% / -10% / 0%
Selling prices	-2% throughout	Recover 40% of incremental material cost	-3% / -1.5% / 0%
Unit material prices	Unchanged	+20%; materials are 50% of base COGS	Unchanged
Additional receivable days	+5 throughout	0	+10 / +5 / 0
Additional inventory days	+10 throughout	+15 throughout	+25 / +12.5 / 0

Supplier payment days remain unchanged. Cash operating expenses are 70% fixed and 30% volume-sensitive. All cases retain base investment, depreciation and debt schedules; no additional savings, investment cuts or financing responses are credited.

2024 material cost recovery

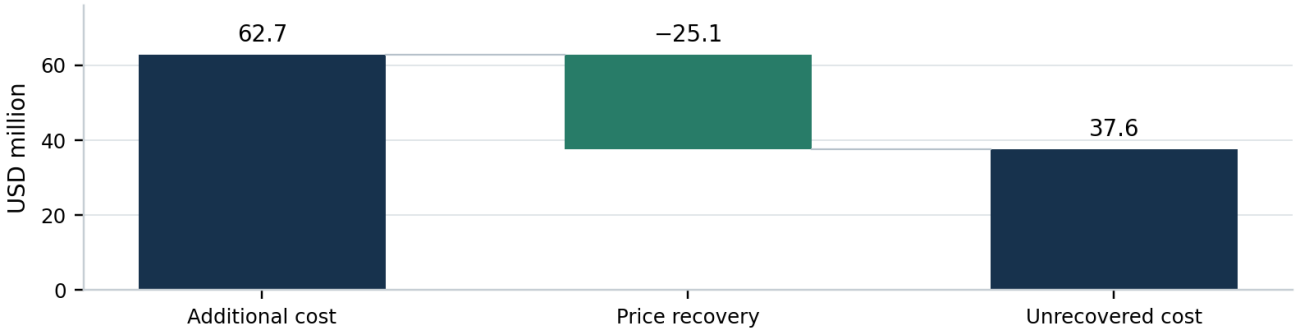


Figure 1. In 2024, only 40% of the additional material cost is recovered through selling prices. All chart values are USD millions unless the axis states otherwise.

The material shock adds **$62.7m** to costs. Pricing recovers **$25.1m**, leaving **$37.6m unrecovered**. Revenue increases 2.4%, but COGS rises 10% versus base. Same-quarter recovery, unchanged volumes and available precautionary stock are simplifying assumptions requiring challenge.

Profitability results

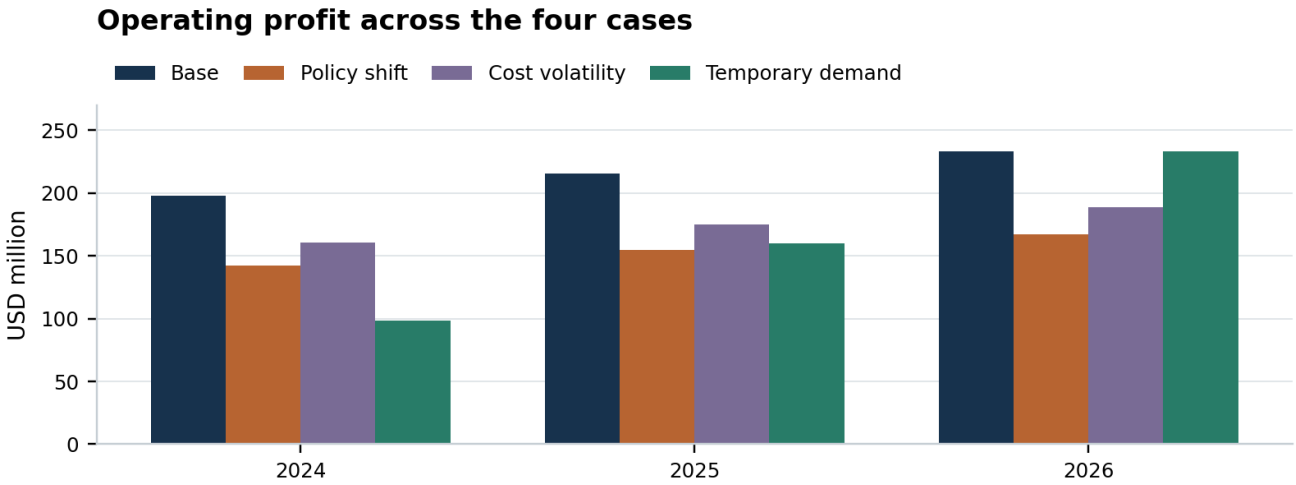


Figure 2. Temporary demand weakness creates the most severe initial operating-profit decline; the assumed recovery closes the earnings gap by 2026.

In the temporary demand case, 2024 revenue falls **22.4%** and operating profit falls **50.2%** versus base, to **$98.6m**. Discounting and lower volumes reduce contribution while most cash overhead remains fixed.

Policy pressure reduces operating profit approximately **28%** versus base across the forecast. Material inflation reduces it **19%**, despite higher nominal sales. Revenue growth therefore does not, by itself, demonstrate stronger financial performance.

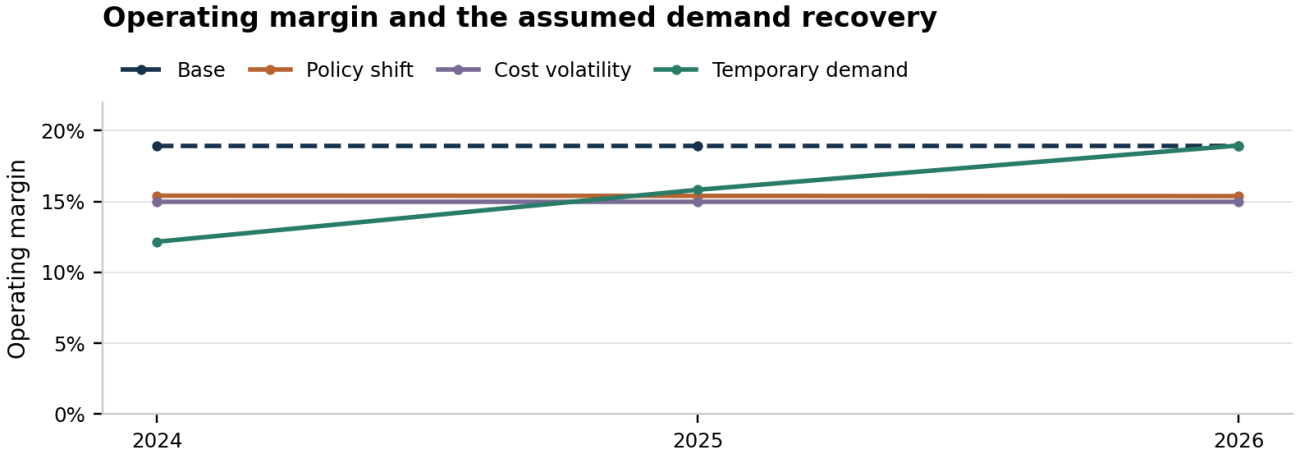


Figure 3. Operating margins distinguish the temporary recovery assumption from continuing policy and material-cost pressure.

Demand-case operating margin recovers from **12.2% in 2024** to the base **18.9% in 2026**. Policy and cost margins remain near **15.4% and 15.0%**. The distinction is persistence: temporary disruption permits an assumed operating recovery, while continuing policy and cost pressure leave lasting margin reductions.

Cash generation and financial risk

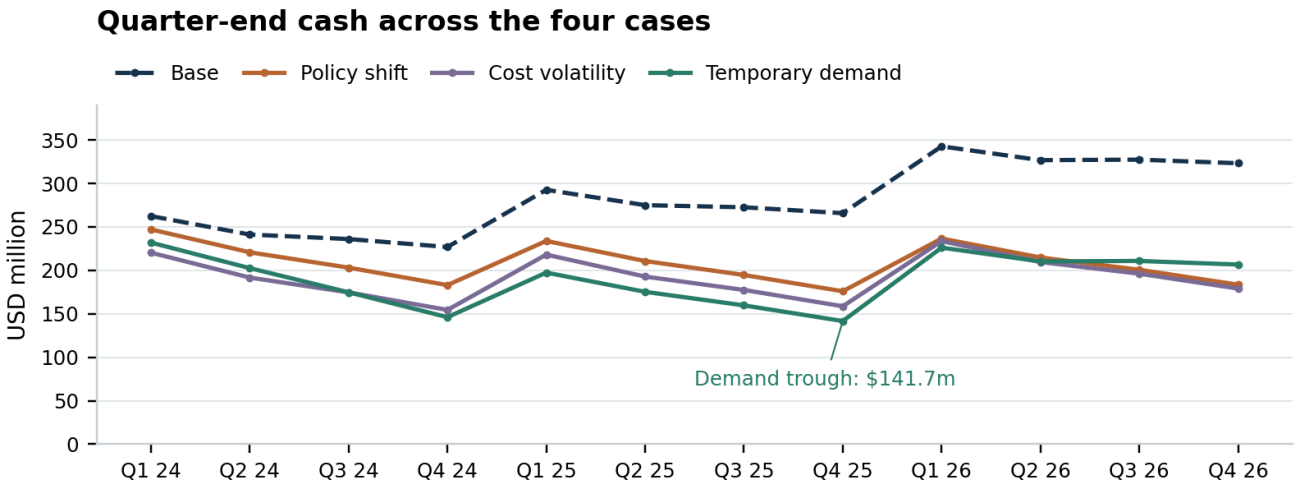


Figure 4. Quarterly cash identifies the timing of financial pressure. The minimum is a forecast quarter-end observation, not minimum daily cash.

Case	Lowest quarter-end cash	Timing	2026 cash
Base	226.8	Q4 2024	323.2
Policy shift	175.9	Q4 2025	183.5
Cost volatility	154.4	Q4 2024	179.0
Temporary demand	141.7	Q4 2025	206.7

The demand case produces the lowest cash trough. Its 2026 operating recovery nevertheless leaves cash **$116.6m below base**, reflecting cumulative earlier earnings losses. Cost volatility leaves the lowest terminal cash because the shock persists.

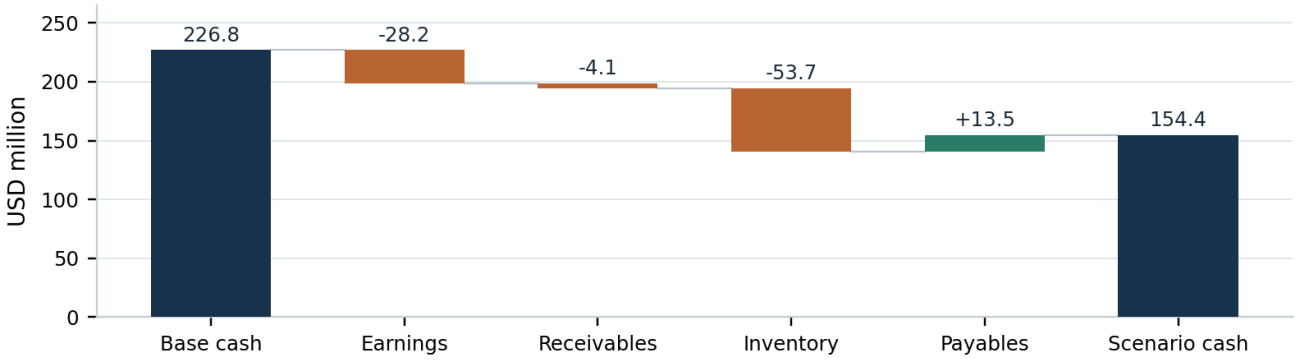


Figure 5. The 2024 cost-case cash bridge separates net-income and working-capital effects. Figures are rounded; totals use unrounded values.

Lower net income, higher inventory and additional receivables reduce cash from **$226.8m to $154.4m**, partly offset by higher payables. Inventory alone absorbs an additional **$53.7m**.

Recommendations

Protect liquidity

Maintain a rolling 13-week cash forecast and define an operating cash buffer from essential payments, maintenance and debt service. Assess financing options before projected cash approaches that buffer.

Control inventory commitments

Link purchasing to orders, lead times and delivery reliability. Challenge uniform precautionary stock increases and compare suppliers on delivered cost, payment terms and reliability.

Defend contribution

Evaluate discounts against product economics and retention. Track actual material-cost recovery and repricing delays instead of assuming the modelled 40% recovery will occur immediately.

Stage investment

Separate maintenance and committed expenditure from discretionary expansion. Preserve capabilities needed for temporary demand recovery while revisiting expansion economics under persistent pressure.

Reforecast from evidence

Monitor orders, realised prices, material costs, overdue receivables and inventory days. Revise the recovery path if demand fails to improve, and revise cost assumptions if customers resist repricing.

Limitations and attribution

Interpretation limits

Debt declines to **$200m by 2026** in every case because scheduled repayments remain funded. Some years use opening cash to finance investment and principal repayments. Positive quarter-end balances do not establish adequate daily liquidity or covenant compliance.

The scenarios are unweighted and have not been combined into a joint stress case. Historical depreciation and gross capex are unavailable; forecast depreciation and asset lives are modelling assumptions, limiting historical-to-forecast cash-flow comparability. COGS is assumed volume-sensitive, with manufacturing fixed costs unidentified. Minimum operating cash requirements, covenant terms and intra-quarter funding needs are unavailable.

Statement reconciliations, independent Python calculations and input-change tests support numerical consistency, not forecast certainty or an independent audit. Native Excel behaviour has not been tested. Recommendations have not been credited with quantified benefits.

Sources and attribution

Source case, historical inputs and broad scenario framework: **Corporate Finance Institute**, Acme Solar Technologies Case Study and accompanying machine-readable data. Model development, quantitative forecast assumptions, scenario calibration and conclusions: **this independent project**, informed by the supplied data. Financial results and charts are drawn from AST_Master_Financial_Model.xlsx. Monetary amounts are USD millions unless stated otherwise.

Use of artificial intelligence

AI tools supported language refinement, financial analysis, coding, model construction and numerical checks. The author directed the project scope and assumptions, reviewed the outputs, and remains responsible for final interpretation and publication. Automated checks do not constitute an independent audit.